# S2 Advanced Statistical Methods for Data Science - Coursework Assignment

Raunaq Rai (rsr45@cam.ac.uk)
Data Intensive Science, Department of Physics, University of Cambridge

04 April, 2025

Word Count: 2982

## Introduction

The Antikythera Mechanism is one of the most remarkable technological discoveries from the ancient world. Found in 1901 (by accident) in a shipwreck near the Greek island of Antikythera, the device dates back to between 140 and 60 BCE (perhaps another opportunity for bayesian inference). What makes it so extraordinary is its system of bronze gears, which could be manually turned to display a wide range of astronomical phenomena. Housed in a wooden box, the mechanism likely showed the positions of the Sun and Moon, lunar phases, eclipse predictions, and even events tied to the ancient Greek calendar such as the Olympic Games [1].

Thanks to X-ray tomography, researchers have reconstructed parts of the mechanism and gained a clearer picture of its workings. One key component is the calendar ring — a circular disk punctured with holes arranged around its edge. Only part of this ring survives, but it provides clues to the type of calendar the device may have used. While it was once thought to represent a 365-day solar calendar, new analysis suggests it may instead follow a 354-day lunar cycle [2].

In this project, we use Bayesian Inference to infer how many holes the full ring originally contained. Using the known positions of 81 holes across eight broken and misaligned fragments, we apply Hamiltonian Monte Carlo Markov Chains to estimate the total number of holes. We use two models for positional uncertainty: one with isotropic Gaussian error, and one separating radial and tangential components[3]. By comparing the results from both models, we aim to determine which best fits the data and what this can tell us about how the mechanism was designed and about the way time was measured in the ancient world.

## 1 Measured Hole Locations $d_i \in \mathbb{R}^2$ in the $x$-$y$ Plane

The dataset [4] contains the $x$ and $y$ coordinates of 81 holes in the surviving fragments of the calendar ring. These holes are distributed across eight fragments, each of which is

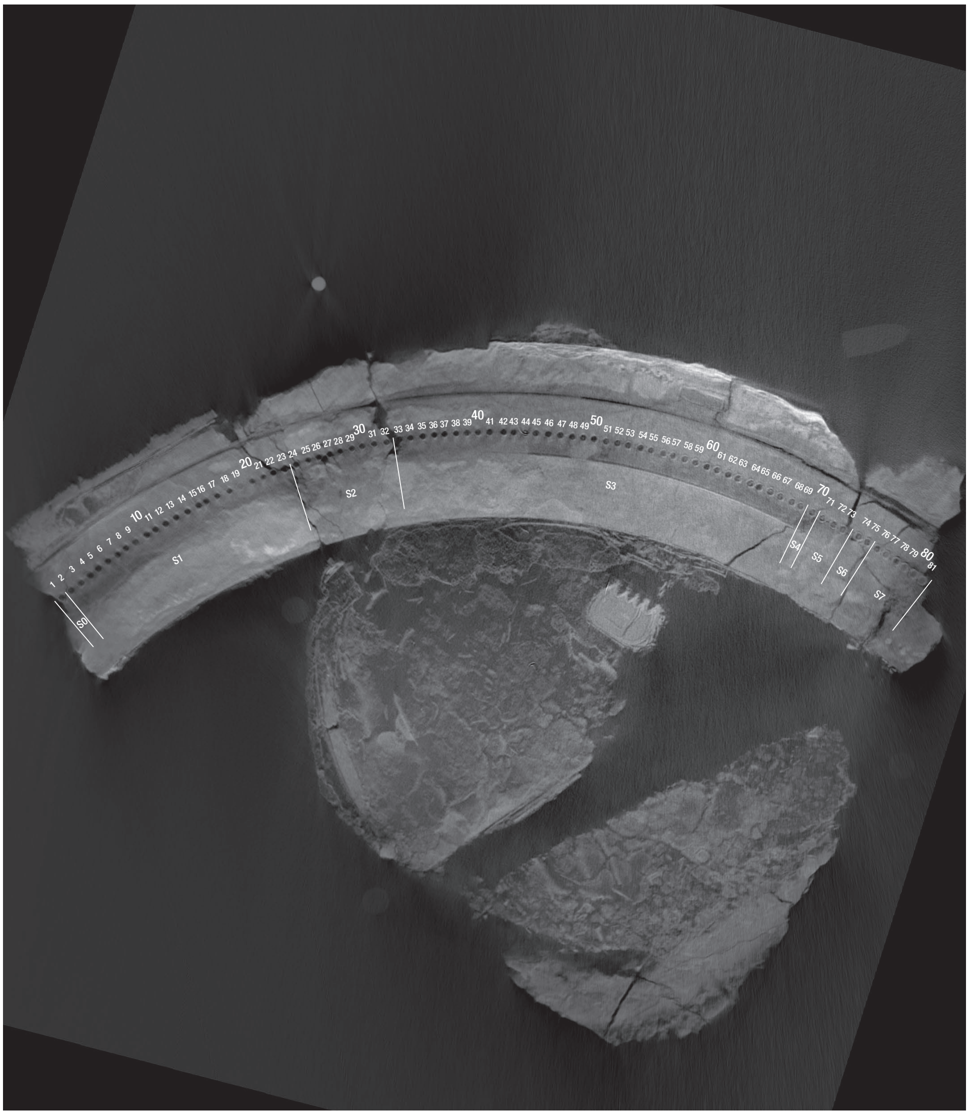



Figure 1: X-ray image of the surviving fragments of the Antikythera Mechanism.

colour-coded in Figure 2.

## 2 Hole Location model

Following the model described in Section 2 of Woan and Bayley [3], we assume that the original calendar ring contained $N$ holes, evenly spaced around a circle of radius $r$. The full ring no longer survives, and the measured hole locations come from 8 fragmented sections. Each fragment is slightly misaligned due to relative translations and in-plane rotations.

$$\vec{m}_i = \begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{pmatrix} r\cos\left(\frac{2\pi(i-1)}{N} + \alpha_j\right) + x_{0j} \\ r\sin\left(\frac{2\pi(i-1)}{N} + \alpha_j\right) + y_{0j} \end{pmatrix}, \tag{1}$$

where $(x_{0j}, y_{0j})$ is the translation of section $j$, and $\alpha_j$ is its relative angular offset. The model assumes all fragments lie in the same $x$-$y$ plane and are internally undistorted.

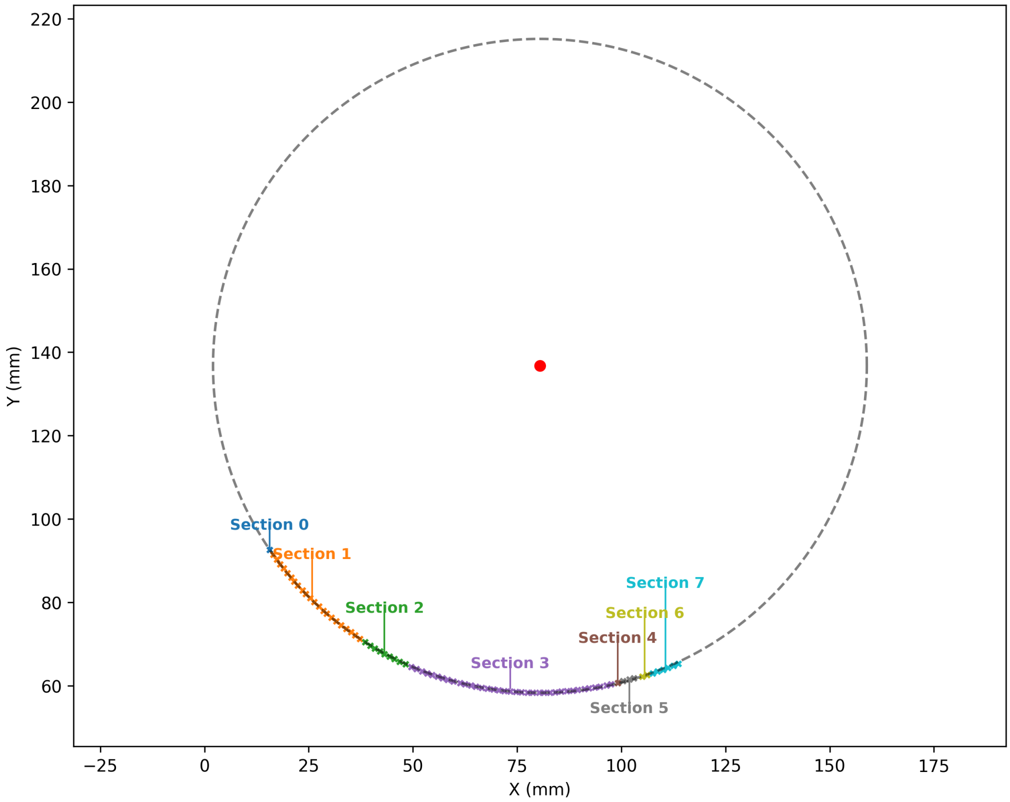


Figure 2: Measured hole locations $d_i \in \mathbb{R}^2$ in the $x$-$y$ plane, colour-coded by fractured ring segment. The dashed circle shows the estimated full ring, fitted to the data using a non-linear least-squares method. Section labels are placed to minimise overlap and connected to the median point of each segment.

## 3 Likelihood Model

To assess how well a proposed model predicts the observed hole positions, we assume that each measured hole location $\vec{d_i} = (x_i, y_i)$ is an independent realisation from a two-dimensional Gaussian distribution centred on a predicted location $\vec{m}_i = (x_i^{\text{model}}, y_i^{\text{model}})$. The set of all measurements is denoted $D = \{\vec{d_i} \mid i = 1, \ldots, N_{\text{holes}}\}$, with $N_{\text{holes}} = 81$.

### 3.1 General Form of the Likelihood

Assuming independent Gaussian errors, the likelihood for a single hole is:

$$p(\vec{d_i} \mid \vec{m}_i, \Sigma) = \frac{1}{2\pi\sqrt{|\Sigma|}} \exp\left(-\frac{1}{2}(\vec{d_i} - \vec{m}_i)^\top \Sigma^{-1} (\vec{d_i} - \vec{m}_i)\right), \tag{2}$$

where $\Sigma$ is the $2 \times 2$ covariance matrix characterising the uncertainty. Aggregating over all holes, the full log-likelihood is:

$$\log \mathcal{L}(D \mid \theta) = -\frac{1}{2} \sum_{i=1}^{N_{\text{holes}}} \left[(\vec{d_i} - \vec{m}_i)^\top \Sigma^{-1} (\vec{d_i} - \vec{m}_i) + \log(2\pi|\Sigma|)\right], \tag{3}$$

where $\theta$ denotes the full set of model parameters:

$$\theta = \{N, r, \text{error parameters in } \Sigma, (x_j, y_j, \alpha_j) \text{ for } j = 0, \ldots, 7\}.$$

## 3.2 Isotropic Gaussian Error Model

In the isotropic case, errors in $x$ and $y$ directions are assumed equal and uncorrelated:

$$\Sigma = \sigma^2 I, \quad \Sigma^{-1} = \frac{1}{\sigma^2} I, \quad |\Sigma| = \sigma^4, \tag{4}$$

with a single noise parameter $\sigma$. Substituting into the log-likelihood yields:

$$\log \mathcal{L}(D \mid \theta) = -\frac{1}{2\sigma^2} \sum_{i=1}^{N_{\text{holes}}} \left\| \vec{d}_i - \vec{m}_i \right\|^2 - \frac{N_{\text{holes}}}{2} \log(2\pi\sigma^2). \tag{5}$$

## 3.3 Anisotropic Radial–Tangential Error Model

To account for direction-dependent uncertainty, we define a local radial–tangential frame for each hole. Given angular position $\phi_i$, the unit vectors are:

$$\hat{r}_i = \begin{pmatrix} \cos\phi_i \\ \sin\phi_i \end{pmatrix}, \qquad \hat{t}_i = \begin{pmatrix} -\sin\phi_i \\ \cos\phi_i \end{pmatrix}. \tag{6}$$

The displacement between predicted and observed positions is:

$$\vec{\delta}_i = \vec{d}_i - \vec{m}_i. \tag{7}$$

Projected onto the local basis:

$$\delta_{r,i} = \hat{r}_i^\top \vec{\delta}_i, \qquad \delta_{t,i} = \hat{t}_i^\top \vec{\delta}_i. \tag{8}$$

#### Mahalanobis Distance and Log-Likelihood

The Mahalanobis distance generalises Euclidean distance by incorporating directional uncertainty:

$$D_i^2 = \begin{pmatrix} \delta_{r,i} \\ \delta_{t,i} \end{pmatrix}^\top \begin{pmatrix} 1/\sigma_r^2 & 0 \\ 0 & 1/\sigma_t^2 \end{pmatrix} \begin{pmatrix} \delta_{r,i} \\ \delta_{t,i} \end{pmatrix} = \frac{\delta_{r,i}^2}{\sigma_r^2} + \frac{\delta_{t,i}^2}{\sigma_t^2}. \tag{9}$$

This distance penalises deviations more strongly in directions of higher precision. It replaces the quadratic form in Equation (2) for the anisotropic case, leading to:

$$\log \mathcal{L}(D \mid \theta) = -\frac{1}{2} \sum_{i=1}^{N_{\text{holes}}} \left[ D_i^2 + \log(2\pi\sigma_r\sigma_t) \right], \tag{10}$$

where the covariance determinant is $|\Sigma_{\text{rt}}| = \sigma_r^2\sigma_t^2$.

## 3.4 Parameter Summary

- $N$ and $r$ define the angular placement and radius of holes around the full ring.
- $(x_j, y_j, \alpha_j)$ describe the translation and rotation of each fractured segment.
- $\sigma$ (isotropic) or $(\sigma_r, \sigma_t)$ (anisotropic) define the scale of measurement uncertainty and influence both the distance and normalisation terms in the log-likelihood.

Both error models were implemented in NumPy and JAX.

# 4 Derivatives of Log-Likelihood

We computed the gradient of the Log-Likelihood with respect to the model parameters $\theta_\mu$:

$$\frac{\partial}{\partial \theta_\mu} \log L(\mathcal{D} \mid \boldsymbol{\theta}) \tag{11}$$

For the derivatives we used automatic differentiation via the jax library. The log-likelihood was implemented in a JAX-compatible format, which allowed for straightforward computation of the gradient and Hessian. Automatic differentiation works by applying the chain rule systematically (as shown in the previous Research Computing C1 coursework) to the sequence of operations used to evaluate the function, allowing exact derivatives to be computed efficiently and accurately.

To ensure the correctness of the computed gradients, we also implemented a finite difference approximation using central differences:

$$\frac{\partial \log L}{\partial \theta_\mu} \approx \frac{\log L(\theta_\mu + \epsilon) - \log L(\theta_\mu - \epsilon)}{2\epsilon} \tag{12}$$

This was compared against the JAX gradients for all parameters. We evaluated both methods at a set of arbitrary parameter values that we selected manually. This allowed us to verify that the automatic differentiation mechanism worked correctly. As an additional check, we computed both forward-mode and reverse-mode gradients using jax's functions, and confirmed consistency across modes.

$$\text{Relative Difference} = \frac{|\nabla_{\text{JAX}} - \nabla_{\text{FD}}|}{|\nabla_{\text{FD}}| + \epsilon} \tag{13}$$

The relative differences were found to be negligible, confirming the accuracy of the automatic differentiation approach.

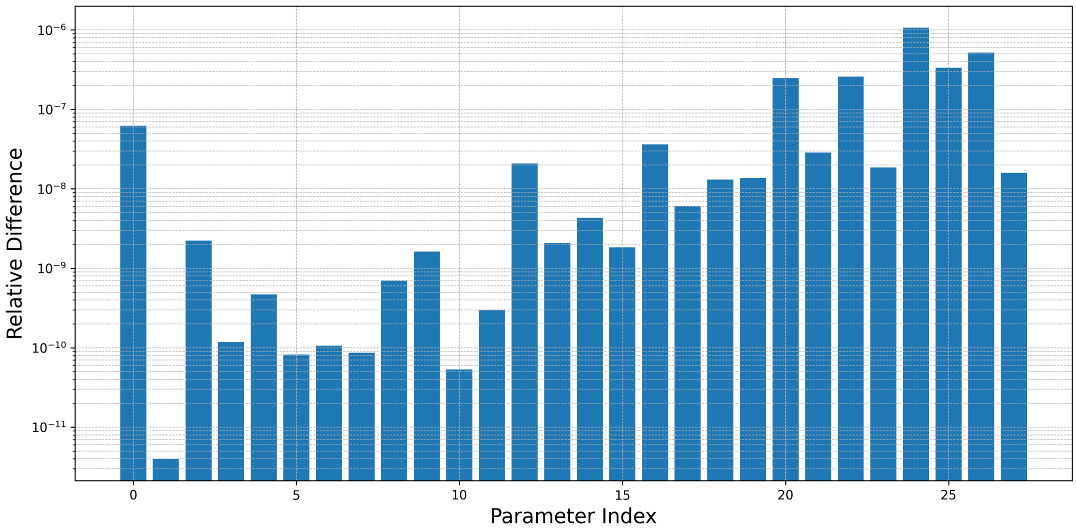



Figure 3: Relative difference between gradients computed via JAX autodiff and finite differences. Each bar corresponds to a parameter $\theta_\mu$.

## 5 Maximum Likelihood Estimation

To estimate the best-fitting model parameters, we performed maximum likelihood estimation (MLE) for both the isotropic and anisotropic covariance models. Sections 0 and 4, which each contained only a single hole, were excluded.

We used the L-BFGS-B optimisation algorithm [5] to maximise the log-likelihood function, with gradients computed via automatic differentiation using jax. Optimisation was carried out on the negative log-likelihood.

The parameter ordering is shown in Table 1.

| Index | Parameter |
|---|---|
| 0 | $N$ (number of holes) |
| 1 | $R$ (ring radius) |
| 2 | $\sigma_r$ (radial uncertainty) |
| 3 | $\sigma_t$ (tangential uncertainty) |
| 4–9 | $\phi_1$ to $\phi_6$ (angular offsets) |
| 10–15 | $x_1$ to $x_6$ (section $x$ offsets) |
| 16–21 | $y_1$ to $y_6$ (section $y$ offsets) |

Table 1: Parameter ordering for the anisotropic model.

Initialisation values were based on the values presented by Woan and Bayley[3] with small perturbations added. Parameter bounds were applied to ensure physical validity (e.g., $N > 0$, $\sigma > 0$).

The optimiser returned the following result:

Maximised Log-Likelihood: 230.67

```
[ 3.55138326e+02  7.73198713e+01  2.57642786e-02  1.22578786e-01
 -2.52567532e+00 -2.13565124e+00 -1.97423759e+00 -1.32147032e+00
 -1.26317302e+00 -1.25305939e+00  7.96772149e+01  7.98973406e+01
  7.98583275e+01  8.13733544e+01  8.15575431e+01  8.32199045e+01
  1.36019135e+02  1.35701360e+02  1.35689755e+02  1.36066927e+02
  1.35837969e+02  1.36401869e+02]
```

Using these parameters, we predicted the expected hole positions for each section and overlaid them on the measured hole locations (Figure 4). To reduce clutter, the predicted points were vertically shifted by 10 mm and arrows were added from the shifted locations to their true positions. The same process was repeated for the isotropic model.

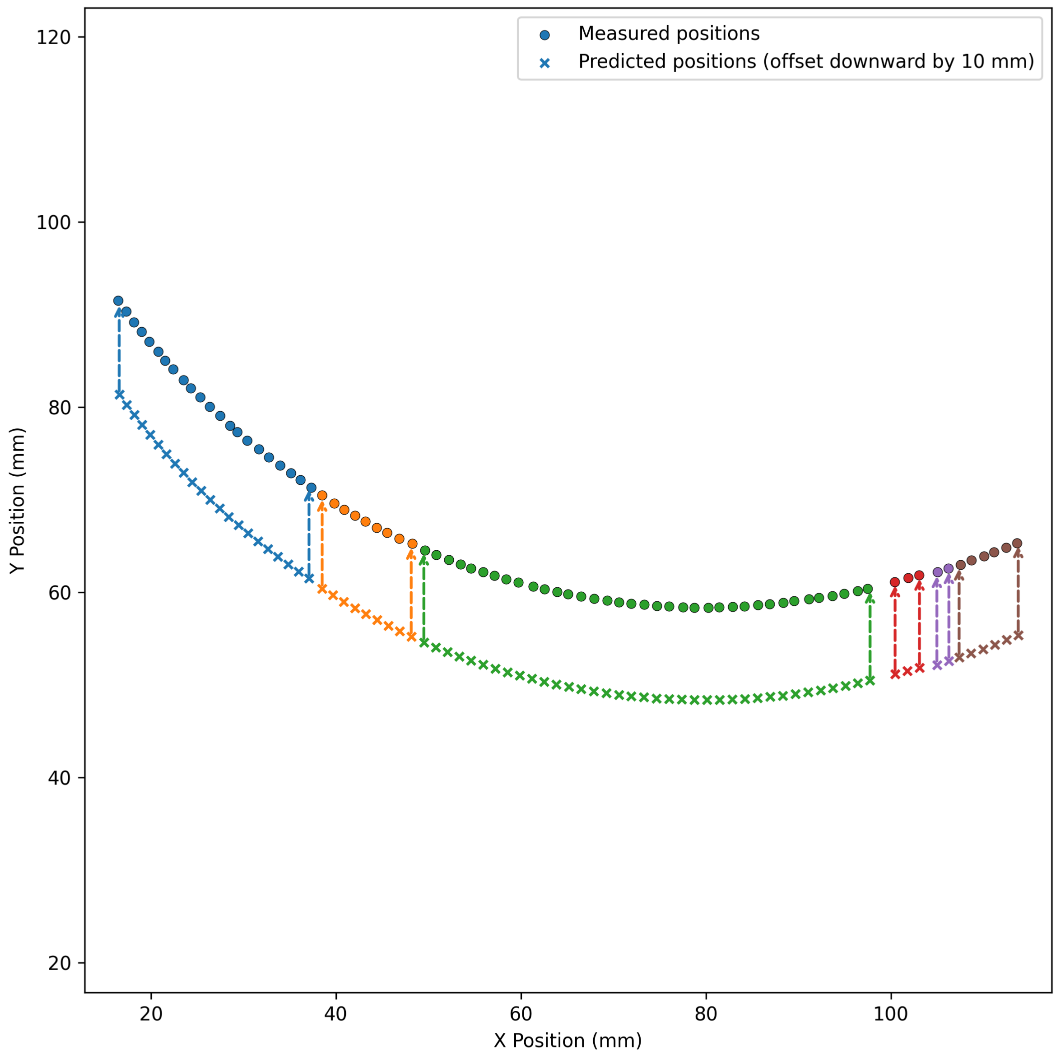



Figure 4: Measured hole positions (circles) overlaid with predicted positions from the anisotropic model (x markers), vertically shifted by 10 mm for clarity. Arrows connect the first and last predicted points in each section to their true model locations. MLE estimation: N = 355.14, R= 77.32.

We repeated the optimisation process for the isotropic model, where a single shared uncertainty parameter $\sigma$ is used for both radial and tangential directions.

The optimiser returned the following result:

Maximised Log-Likelihood: $-20.37$

```
[ 3.54953104e+02  7.72779993e+01  1.24925618e-01 -2.52545406e+00
 -2.13497172e+00 -1.97450722e+00 -1.31829739e+00 -1.26683207e+00
 -1.25314722e+00 7.96396936e+01  7.98280318e+01  7.98622123e+01
  8.11471212e+01  8.18393995e+01  8.32405652e+01  1.36016332e+02
  1.35689853e+02  1.35638898e+02  1.35961123e+02  1.35884849e+02
  1.36361957e+02]
```

As before, we predicted the hole positions using the optimised parameters and overlaid them on the measured data (Figure 5).

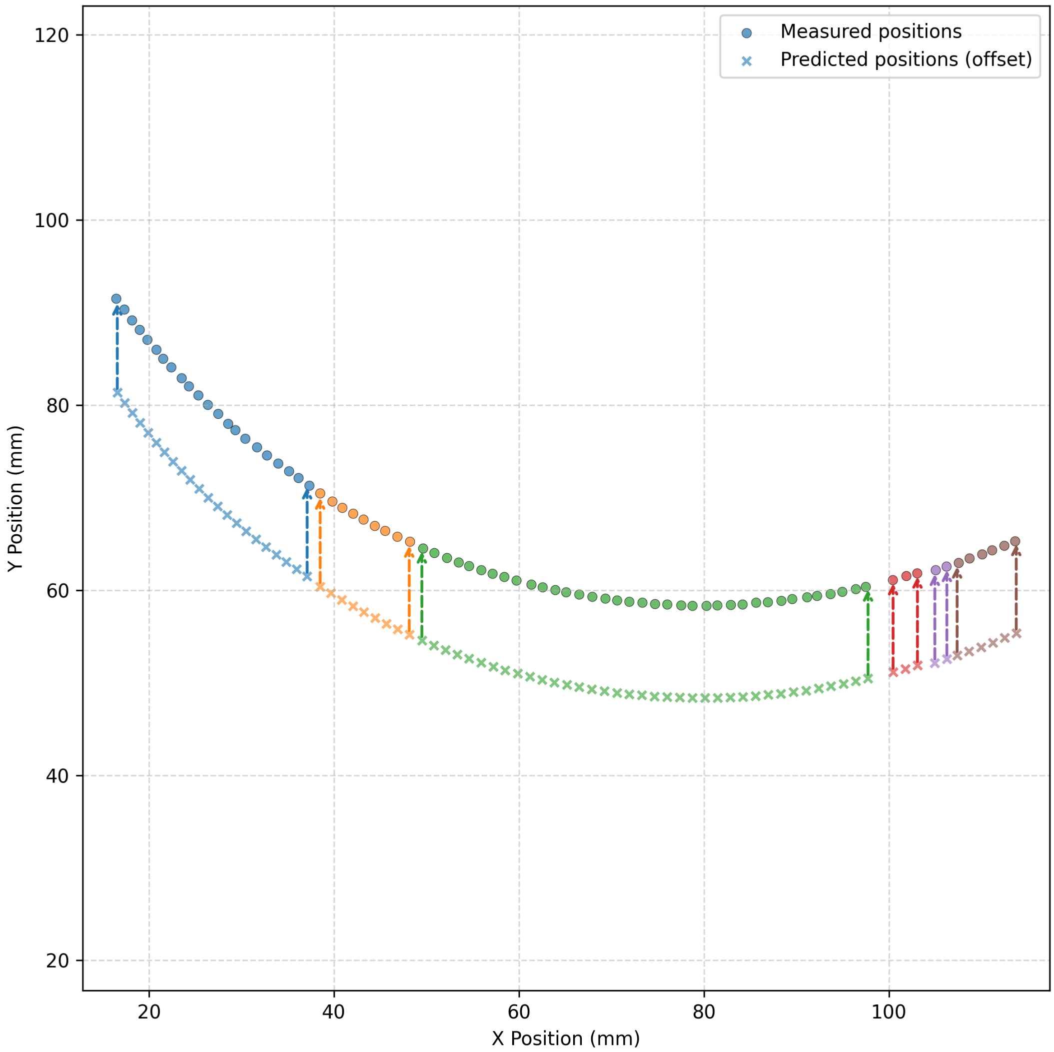



Figure 5: Measured hole positions (circles) overlaid with predicted positions from the isotropic model (x markers), vertically shifted by 10 mm for clarity. Arrows connect the first and last predicted points in each section to their true model locations. MLE estimation: N = 354.95, R= 77.38.

Comparing the two models, the anisotropic version achieved a higher maximised log-likelihood ($\log\mathcal{L} = 230.67$) compared to the isotropic model ($\log\mathcal{L} = -20.37$). This difference suggests that allowing separate treatment of radial and tangential uncertainties provides a better fit to the measured hole positions.

## 6 Bayesian Inference and Posterior Sampling

We performed Bayesian inference using Hamiltonian Monte Carlo (HMC) sampling[6]. This allowed us to estimate the full posterior distributions of the model parameters, incorporating prior knowledge and the observed data.

HMC works by simulating a particle moving through the parameter space under a fictional physical system, using gradients of the log-posterior to propose distant, high-probability samples efficiently and with reduced random walk behaviour compared to traditional MCMC methods.

The priors were informed by previous literature, physical considerations and the hypothesis that the mechanism tracks a lunar calendar. These priors are summarised below and visualised in Figure 6:

- Total number of holes: $N \sim \mathcal{N}(354.4, 3.5)$, reflecting the hypothesis that the calendar ring encodes a lunar year of 12 months (each $\approx 29.53$ days). The mean is informed by our own maximum likelihood estimate (MLE), as well as historical interpretations of the mechanism's purpose as a calendrical device [7][8][9].
- Ring radius: $R \sim \mathcal{N}(77.35, 2)$, based on the MLE results and results published by Woan and Bayley [3].
- Uncertainties:
  - For the anisotropic model, radial and tangential uncertainties were given log-normal priors to reflect the fact that uncertainties cannot be negative. The peaks of these priors also align with results from our MLE and results published by Woan and Bayley [3].

$$\sigma_r \sim \text{LogNormal}(\log(0.028), 0.25), \quad \sigma_t \sim \text{LogNormal}(\log(0.13), 0.3).$$

  - For the isotropic model, the shared uncertainty parameter was:

$$\sigma \sim \text{LogNormal}(\log(0.13), 0.45).$$

- Section-specific parameters: angular offsets $\phi_j$, and spatial offsets $x_j$, $y_j$. Were chosen to align with our MLE results as well as results published by Woan and Bayley [3]. These were given a large standard deviation to allow our HMC to explore a broad parameter space looking at within-section misalignments and minor translational shifts between fractured segments:

$$\phi_j \sim \mathcal{N}(-2.5, 0.5), \quad x_j \sim \mathcal{N}(80, 5), \quad y_j \sim \mathcal{N}(135, 5).$$

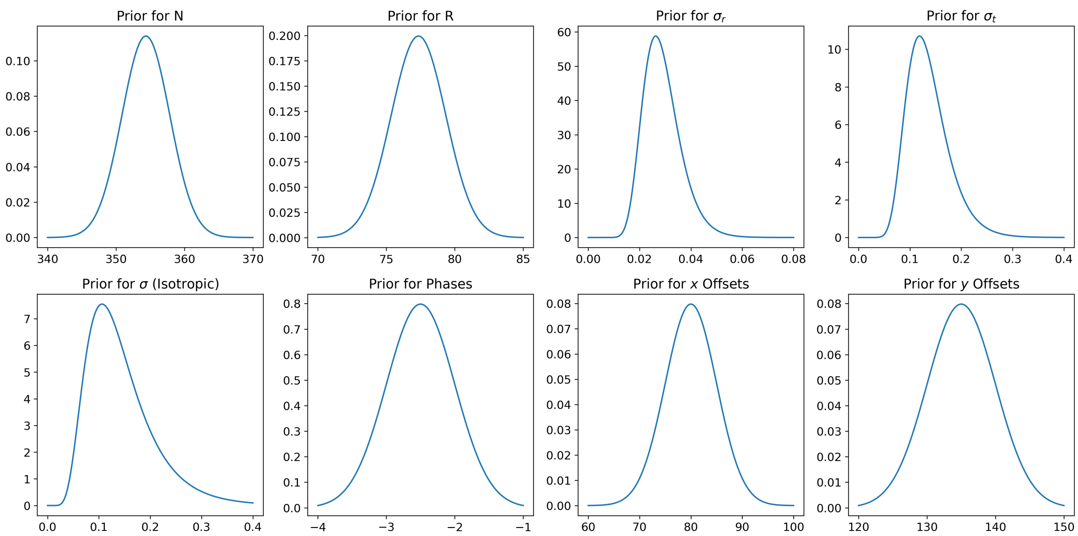



Figure 6: Visualisation of the prior distributions for the model parameters. Each subplot corresponds to one of the priors described above.

Posterior sampling was performed using the No-U-Turn Sampler [10], a variant of HMC known for its efficiency in high-dimensional spaces. We used 12 parallel chains (corresponding to the number of CPU cores on my machine), each with 10,000 warmup steps and 50,000 sampling steps. Initialisation was done via the 'init_to_median' strategy which initialises each parameter based on the median of its prior distribution. This provides a stable and reasonable starting point for the sampler helping ensure faster convergence during warm-up, particularly useful in high-dimensional spaces.

## Anisotropic Model Results

For the anisotropic model, trace plots for key parameters (Figure 7) show good mixing and convergence across all chains, with minimal autocorrelation. The posterior distributions and joint correlations between parameters are shown in the corner plot in Figure 8. Detailed estimates and statistics from this model are shown in the appendix in Table 2.

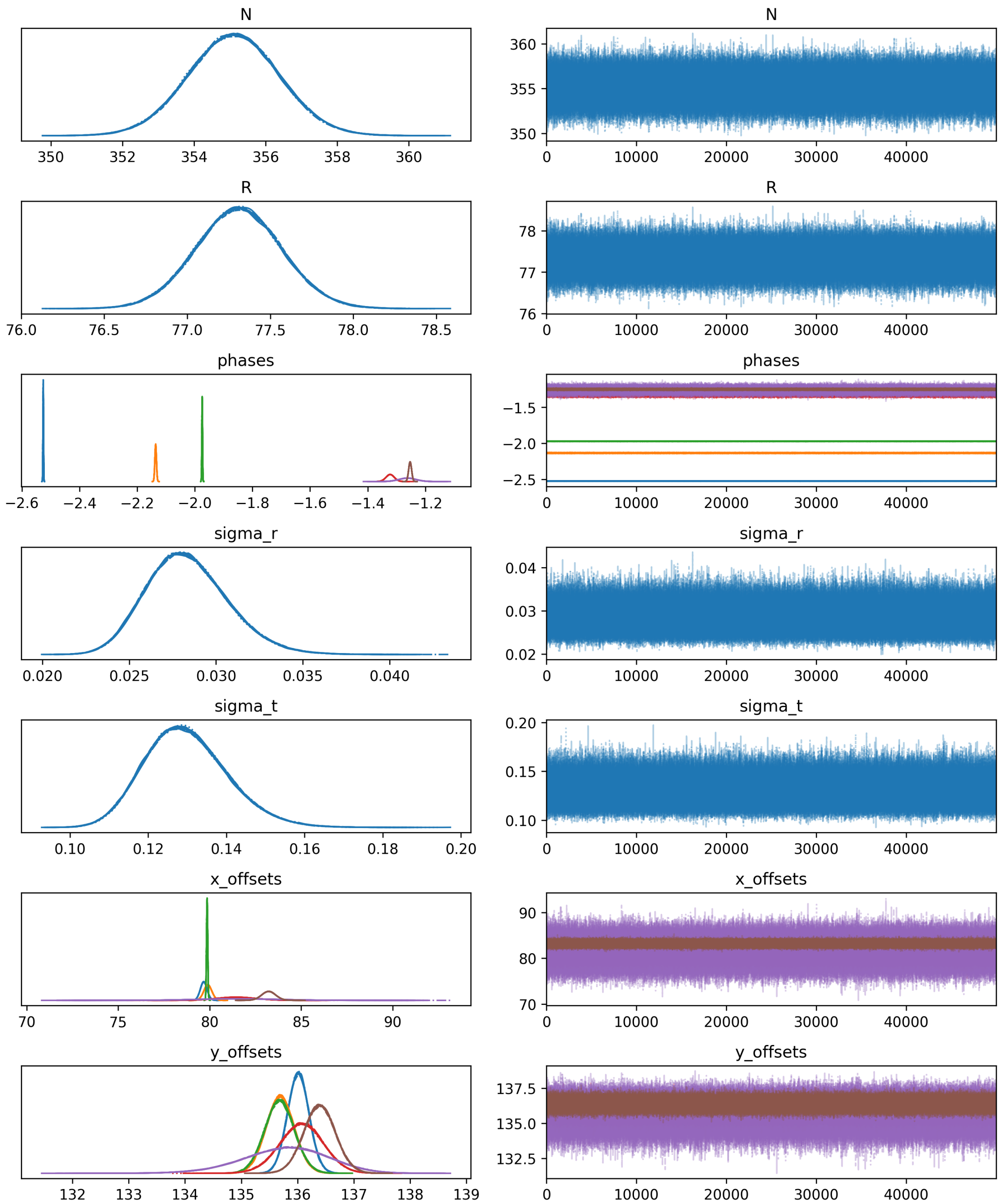



Figure 7: Trace plots for the anisotropic model across all 12 chains

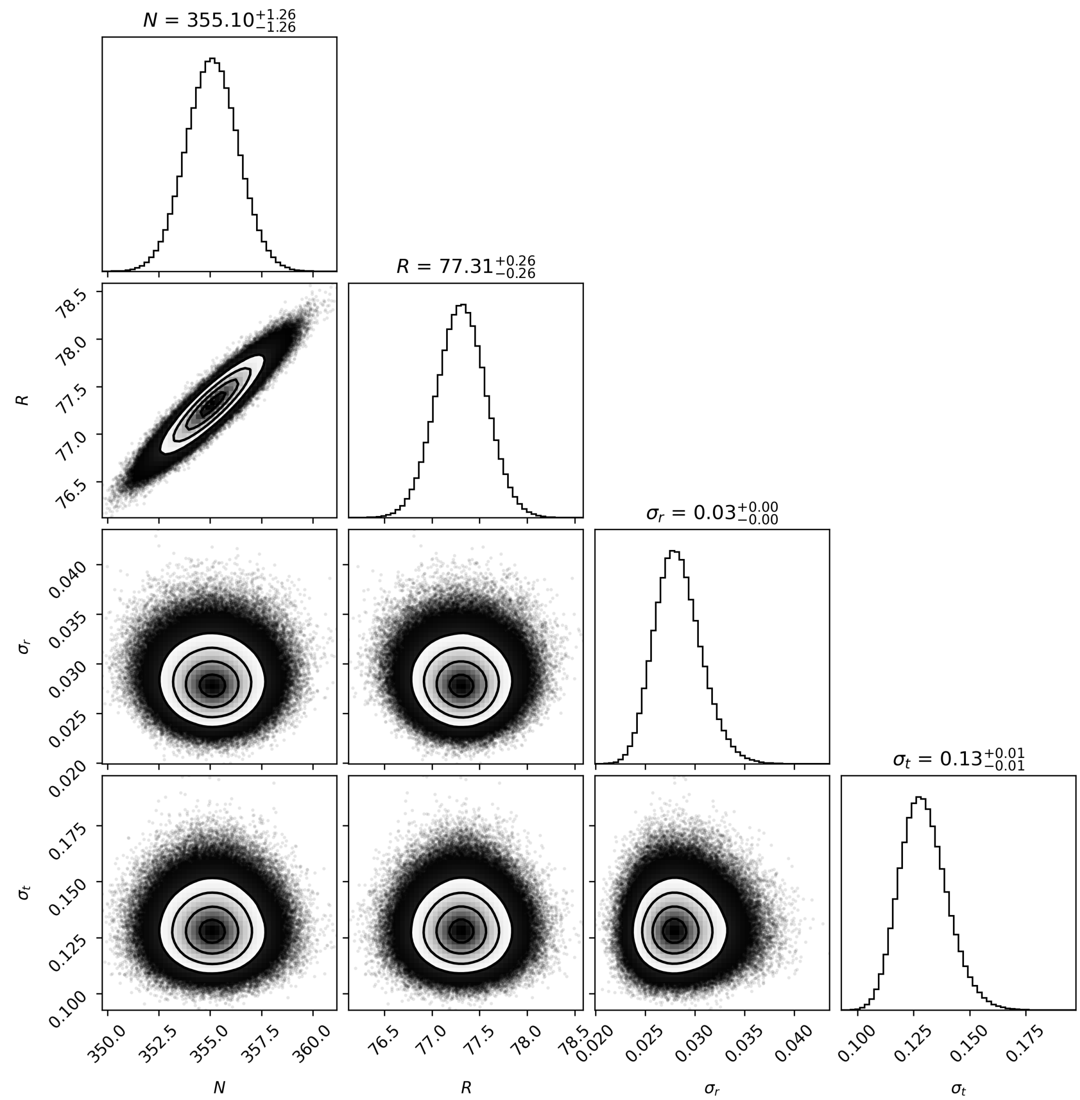



Figure 8: Corner plot showing posterior distributions for the anisotropic model parameters.

## Isotropic Model Results

The isotropic model was fit using the same MCMC settings. The trace plots (Figure 9) again demonstrate good convergence, and the corresponding corner plot (Figure 10) summarises the posterior distributions and joint parameter behaviour. Detailed estimates and statistics from this model are shown in the appendix in Table 3.

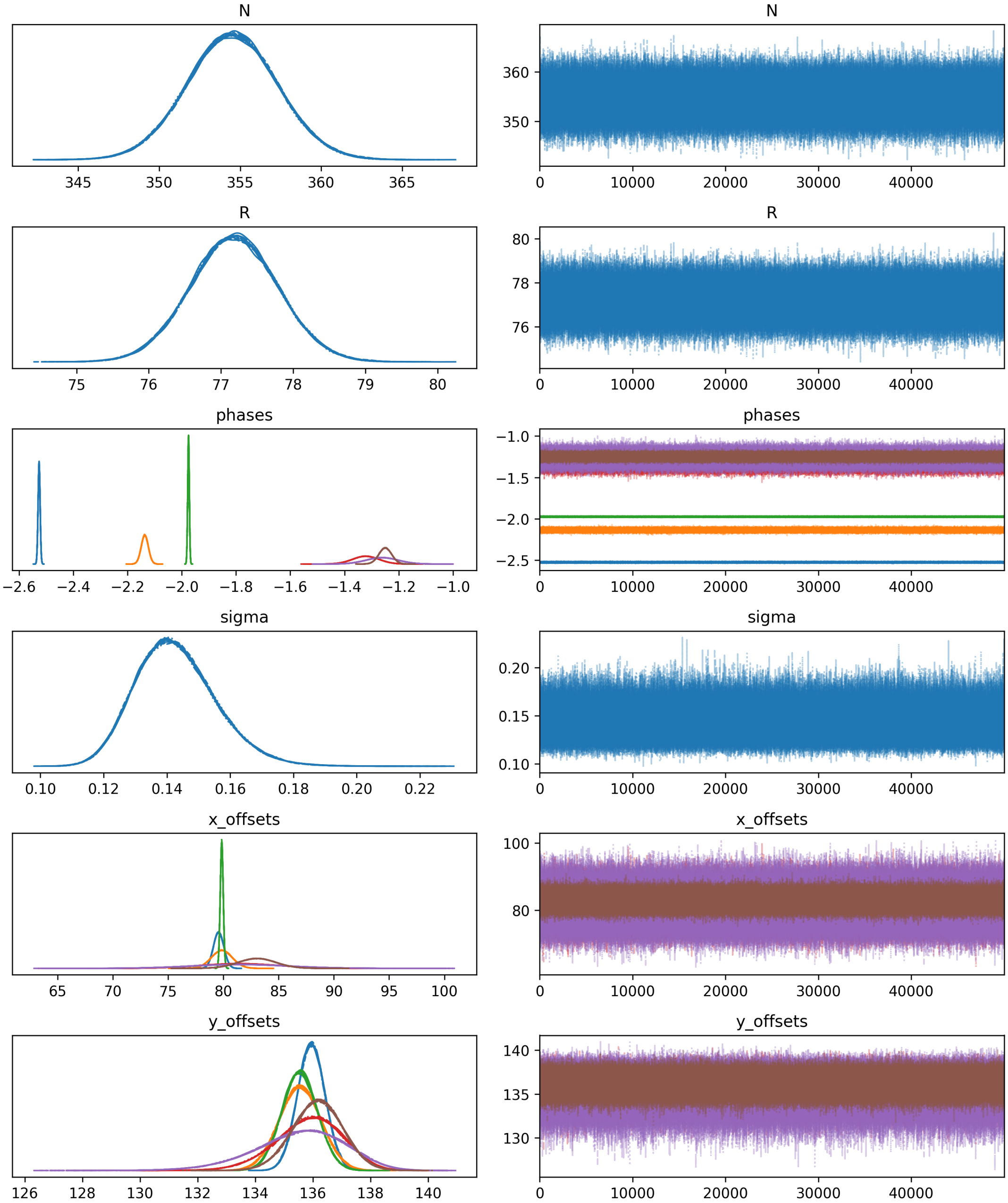


Figure 9: Trace plots for the isotropic model parameters across 12 chains.

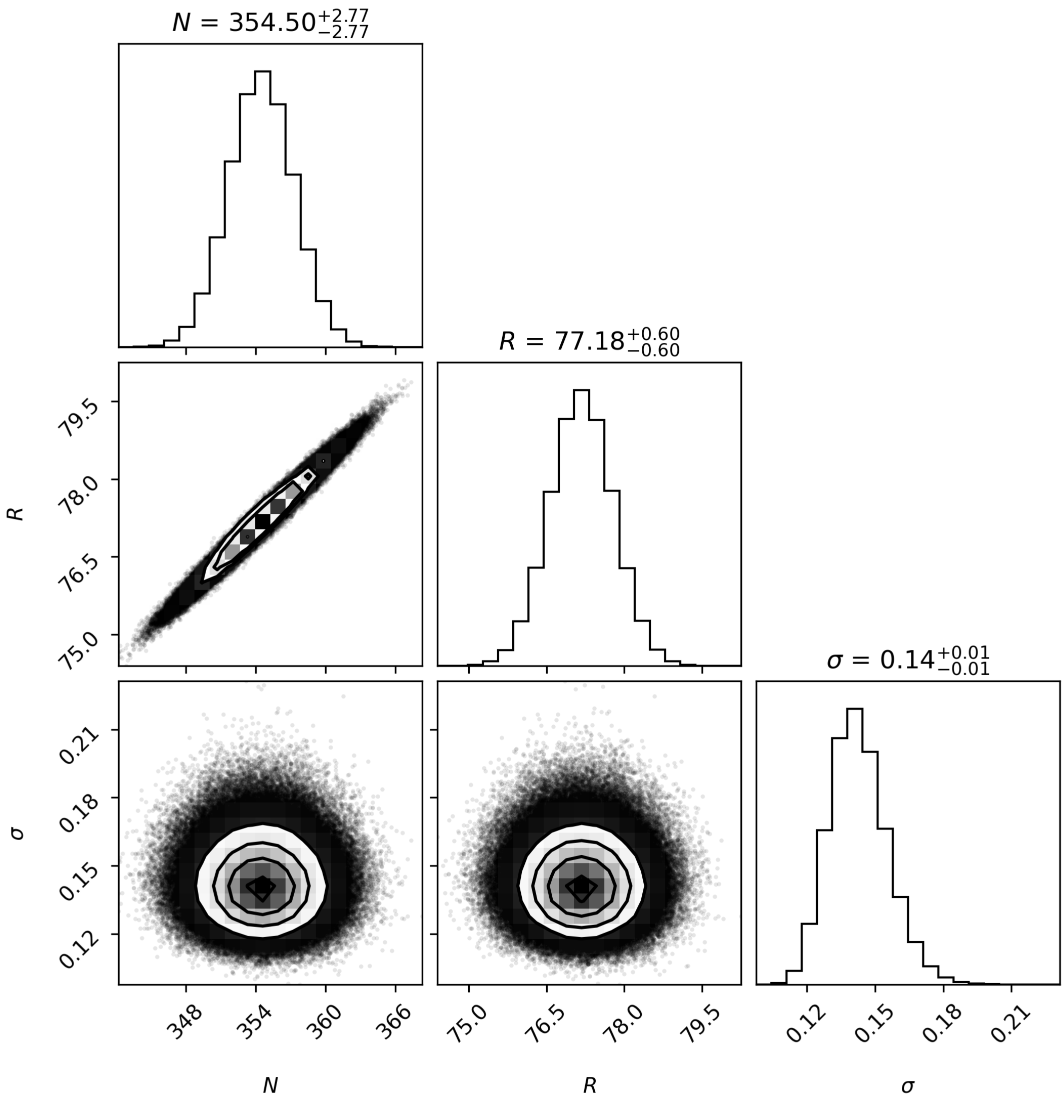
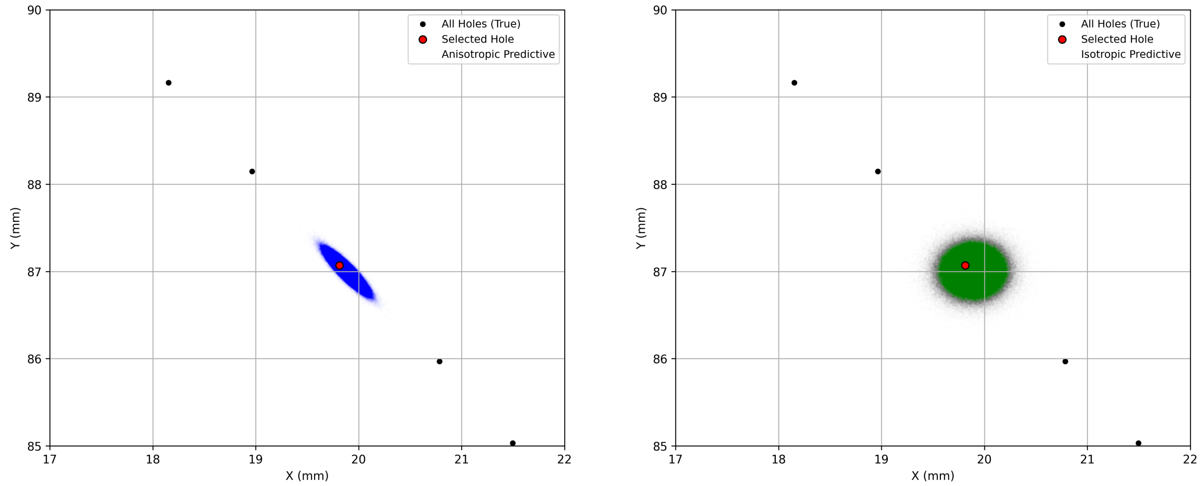



Figure 10: Corner plot showing the posterior distributions for the isotropic model parameters.




(a) Anisotropic model (b) Isotropic model

Figure 11: Posterior predictive distributions for the position of a single hole (Section 1, Hole 5) under the anisotropic and isotropic models. Black dots show all holes in the selected section; the red dot marks the true location of the selected hole. Blue and green dots represent posterior predictive samples under each model.

The predictive distributions show clear differences between the two models. In the isotropic case (Figure 11b), the posterior predictive samples form a near-circular cloud around the true location, which is consistent with the assumption of equal measurement uncertainty in all directions.

The anisotropic model (Figure 11a) produces an elliptical distribution. This shape reflects the assumption that uncertainty differs between the radial and tangential directions. Radial positions are better constrained due to the curvature of the ring, while tangential offsets are more sensitive to misalignments between broken fragments. This results in a tight ellipse along the radial direction and a broader spread tangentially.

The sharpness and clarity of these distributions are a result of the large number of posterior samples used: 600,000 raw samples were produced.

## 7 Covariance Structure and Model Comparison

In two dimensions, the covariance matrix $\Sigma$ defines the shape and orientation of the uncertainty ellipse around each predicted hole location $\vec{m}_i$. In the isotropic case, the same uncertainty is assumed in all directions, leading to circular error regions:

$$\Sigma = \sigma^2 I.$$

However, this may be overly simplistic. A more physically realistic model accounts for different uncertainty along the radial and tangential directions of the ring:

$$\Sigma_{\text{rt}} = \begin{pmatrix} \sigma_r^2 & 0 \\ 0 & \sigma_t^2 \end{pmatrix}$$

This anisotropic model reflects the idea that measurement uncertainty is typically larger in the direction around the ring (tangential) than towards/away from the centre (radial), especially when holes are arc-fitted or distorted by curvature.

## Model Comparison

We used the Savage-Dickey density ratio to compute the Bayes factor between the isotropic and anisotropic models. This method is valid when one model (the isotropic model) is a special case of the other (anisotropic model), obtained by setting a specific parameter to a fixed value.

The isotropic model corresponds to $\Delta = \sigma_t - \sigma_r = 0$, where the radial and tangential uncertainties are equal. The Savage-Dickey ratio compares the posterior and prior densities of this parameter at $\Delta = 0$:

$$\text{Bayes Factor} = \frac{p(\Delta = 0 \mid D)}{p(\Delta = 0)}.$$

We estimated these densities using kernel density estimation applied to posterior and prior samples of $\Delta$. The posterior placed almost no probability mass at $\Delta = 0$, whereas the prior allowed moderate support:

$$\text{Posterior density at } \Delta = 0: \quad 0.00000, \quad \text{Prior: } 0.00868,$$

$$\Rightarrow \text{Bayes Factor (isotropic vs. anisotropic)} \approx 0.000.$$

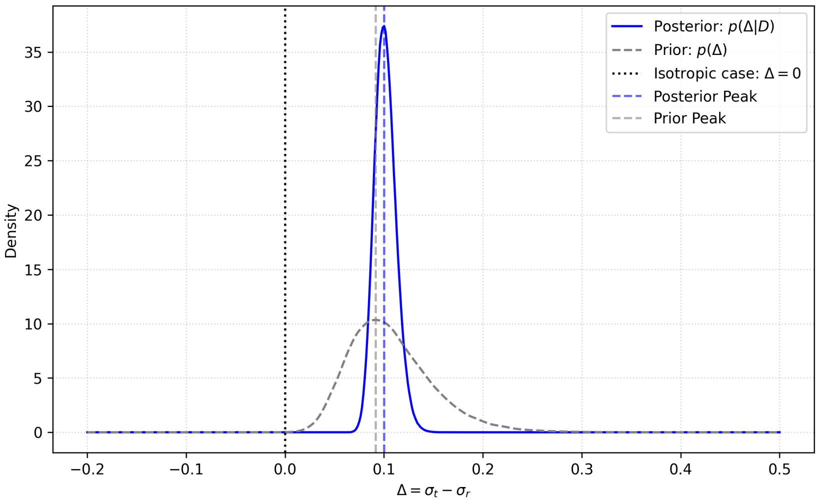



Figure 12: Savage-Dickey density ratio comparing prior and posterior densities of $\Delta = \sigma_t - \sigma_r$.

The posterior distribution is sharply peaked away from zero, result is evidence against the isotropic model. The data favours a model in which radial and tangential uncertainties differ. The radial–tangential error structure likely arises from the way the ring was measured or manufactured, with uncertainty elongated along the arc.

## Nested Sampling and Log-Evidence Comparison

We used nested sampling to estimate the evidence for each model. Unlike the Savage-Dickey ratio, which only compares posterior and prior densities at a single point, nested sampling provides a measure of how well a model explains the data across its entire parameter space.

The evidence for a model $\mathcal{M}$ is defined as:

$$\log Z = \log \int \mathcal{L}(\theta)\, \pi(\theta)\, d\theta,$$

where $\mathcal{L}(\theta)$ is the likelihood and $\pi(\theta)$ is the prior. A greater evidence indicates a better balance between goodness-of-fit and model complexity.

Nested sampling draws a set of live points from the prior and iteratively replaces the worst-fitting point with a better one, gradually focusing on regions of high likelihood. This allows both the evidence and the posterior to be estimated simultaneously.

In our implementation, we used the dynesty library with 500 live points and the 'rwalk' sampling method. The log-likelihood function was compiled using `jax.jit` for speed, and the prior transform was matched to the priors used in our Bayesian inference.

The resulting log-evidence estimates were:

$$\log Z_{\text{aniso}} = 126.75 \pm 0.72, \quad \log Z_{\text{iso}} = -111.94 \pm 0.65.$$

The difference in log-evidence,

$$\Delta \log Z = \log Z_{\text{aniso}} - \log Z_{\text{iso}} = 238.69 \pm 0.97,$$

is in favour of the anisotropic model: a log-evidence differenceabove 100 is decisive evidence for the favourability of the Anisotropic model. Together with the Savage-Dickey result, this confirms that the data clearly favour a model that allows for separate treatment of radial and tangential uncertainties.

## Conclusion

The anisotropic model consistently outperformed the isotropic model. It provided more accurate posterior predictive distributions for individual hole positions, capturing elliptical error structures consistent with physical intuition. Quantitatively, the Savage-Dickey ratio yielded a Bayes factor near zero in favour of the isotropic model. Furthermore, nested sampling revealed a substantial log-evidence difference of $\Delta \log Z \approx 268$ in favour of the anisotropic model.

Beyond statistical performance, the anisotropic model offers a more realistic representation of the measurement process - while radial positions are relatively well constrained by the curvature of the ring, tangential positions are more susceptible to misalignment and fragment distortion. This is reflected in the recovered posterior: $\sigma_r < \sigma_t$ with high confidence.

In summary, our Anisotropic HMC analysis yields a total hole count of $355.1 \pm 1.26$, with a 90% credible interval ranging from 352.98 to 357.16 which is consistent with a 354-day lunar calendar.

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## Appendix: Posterior Summary of Model Parameters

## Anisotropic Model Posterior Summary

| Parameter | Mean | Std | Median | 5% | 95% | n_eff | $\hat{R}$ |
|---|---|---|---|---|---|---|---|
| N | 355.10 | 1.27 | 355.10 | 352.98 | 357.17 | 150089.40 | 1.00 |
| R | 77.31 | 0.26 | 77.31 | 76.89 | 77.74 | 140073.39 | 1.00 |
| phases[0] | -2.53 | 0.00 | -2.53 | -2.53 | -2.52 | 266995.91 | 1.00 |
| phases[1] | -2.14 | 0.00 | -2.14 | -2.14 | -2.13 | 549871.36 | 1.00 |
| phases[2] | -1.97 | 0.00 | -1.97 | -1.98 | -1.97 | 169830.53 | 1.00 |
| phases[3] | -1.32 | 0.01 | -1.32 | -1.35 | -1.30 | 407464.84 | 1.00 |
| phases[4] | -1.26 | 0.03 | -1.26 | -1.31 | -1.21 | 376666.94 | 1.00 |
| phases[5] | -1.25 | 0.01 | -1.25 | -1.26 | -1.24 | 488787.59 | 1.00 |
| $\sigma_r$ | 0.03 | 0.00 | 0.03 | 0.02 | 0.03 | 566747.07 | 1.00 |
| $\sigma_t$ | 0.13 | 0.01 | 0.13 | 0.11 | 0.15 | 624075.38 | 1.00 |
| x_offsets[0] | 79.67 | 0.19 | 79.67 | 79.36 | 79.97 | 146941.56 | 1.00 |
| x_offsets[1] | 79.90 | 0.22 | 79.90 | 79.54 | 80.27 | 287612.22 | 1.00 |
| x_offsets[2] | 79.86 | 0.03 | 79.86 | 79.80 | 79.91 | 242810.23 | 1.00 |
| x_offsets[3] | 81.44 | 1.08 | 81.44 | 79.69 | 83.23 | 411059.14 | 1.00 |
| x_offsets[4] | 81.51 | 2.16 | 81.50 | 77.97 | 85.07 | 378157.08 | 1.00 |
| x_offsets[5] | 83.22 | 0.38 | 83.22 | 82.59 | 83.85 | 443628.60 | 1.00 |
| y_offsets[0] | 136.01 | 0.19 | 136.01 | 135.70 | 136.33 | 142977.35 | 1.00 |
| y_offsets[1] | 135.69 | 0.25 | 135.69 | 135.28 | 136.10 | 159439.48 | 1.00 |
| y_offsets[2] | 135.68 | 0.26 | 135.68 | 135.25 | 136.12 | 140196.16 | 1.00 |
| y_offsets[3] | 136.07 | 0.39 | 136.07 | 135.44 | 136.71 | 210488.10 | 1.00 |
| y_offsets[4] | 135.78 | 0.75 | 135.81 | 134.57 | 137.03 | 303370.16 | 1.00 |
| y_offsets[5] | 136.39 | 0.28 | 136.39 | 135.93 | 136.85 | 167609.76 | 1.00 |

Table 2: Posterior summary statistics for the anisotropic model parameters.

## Isotropic Model Posterior Summary

| Parameter | Mean | Std | Median | 5% | 95% | n_eff | $\hat{R}$ |
|---|---|---|---|---|---|---|---|
| N | 354.50 | 2.79 | 354.50 | 349.91 | 359.11 | 189887.40 | 1.00 |
| R | 77.18 | 0.61 | 77.18 | 76.17 | 78.17 | 187735.89 | 1.00 |
| phases[0] | -2.53 | 0.00 | -2.53 | -2.53 | -2.52 | 437517.22 | 1.00 |
| phases[1] | -2.14 | 0.01 | -2.14 | -2.16 | -2.11 | 387101.74 | 1.00 |
| phases[2] | -1.97 | 0.00 | -1.97 | -1.98 | -1.97 | 236700.53 | 1.00 |
| phases[3] | -1.32 | 0.05 | -1.32 | -1.40 | -1.24 | 357285.89 | 1.00 |
| phases[4] | -1.26 | 0.06 | -1.26 | -1.36 | -1.16 | 339319.30 | 1.00 |
| phases[5] | -1.25 | 0.02 | -1.25 | -1.29 | -1.21 | 377427.05 | 1.00 |
| $\sigma$ | 0.14 | 0.01 | 0.14 | 0.12 | 0.16 | 448359.76 | 1.00 |
| x_offsets[0] | 79.57 | 0.47 | 79.57 | 78.80 | 80.34 | 221776.87 | 1.00 |
| x_offsets[1] | 79.86 | 0.95 | 79.86 | 78.31 | 81.42 | 366747.92 | 1.00 |
| x_offsets[2] | 79.85 | 0.14 | 79.85 | 79.63 | 80.08 | 481017.58 | 1.00 |
| x_offsets[3] | 81.46 | 3.62 | 81.44 | 75.50 | 87.41 | 359572.52 | 1.00 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| x_offsets[4] | 81.53 | 4.35 | 81.49 | 74.44 | 88.78 | 341364.77 | 1.00 |
| x_offsets[5] | 83.04 | 1.72 | 83.04 | 80.18 | 85.83 | 378210.55 | 1.00 |
| y_offsets[0] | 135.94 | 0.48 | 135.94 | 135.16 | 136.73 | 203864.59 | 1.00 |
| y_offsets[1] | 135.55 | 0.72 | 135.55 | 134.37 | 136.75 | 235688.08 | 1.00 |
| y_offsets[2] | 135.54 | 0.62 | 135.54 | 134.52 | 136.55 | 187794.31 | 1.00 |
| y_offsets[3] | 135.85 | 1.16 | 135.91 | 133.99 | 137.77 | 269183.78 | 1.00 |
| y_offsets[4] | 135.54 | 1.55 | 135.65 | 133.05 | 138.09 | 288180.62 | 1.00 |
| y_offsets[5] | 136.16 | 0.87 | 136.17 | 134.75 | 137.60 | 248778.41 | 1.00 |

Table 3: Posterior summary statistics for the isotropic model parameters.

## Corner Plots from Nested Sampling

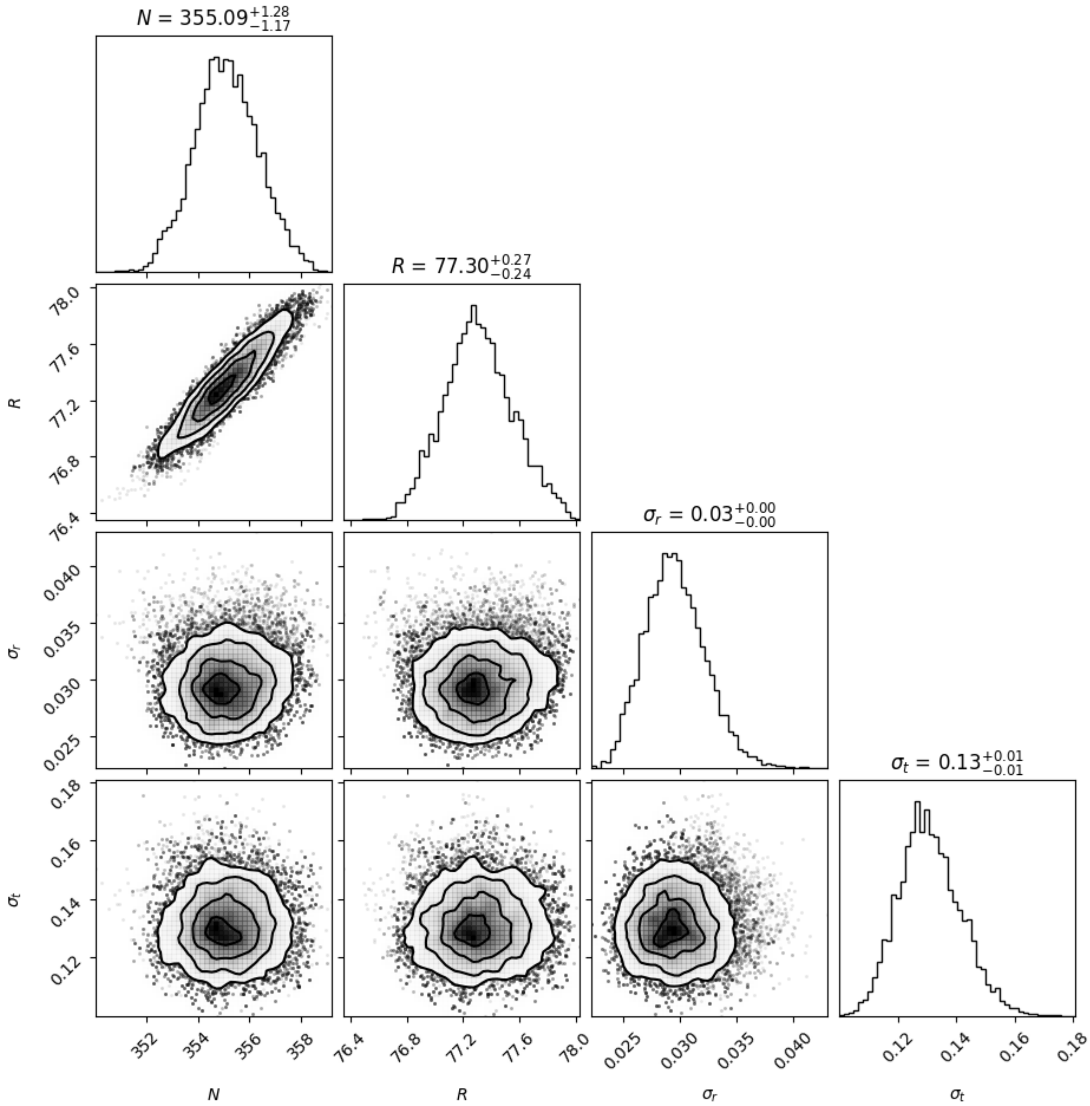


Figure 13: Corner plot from nested sampling for the anisotropic model

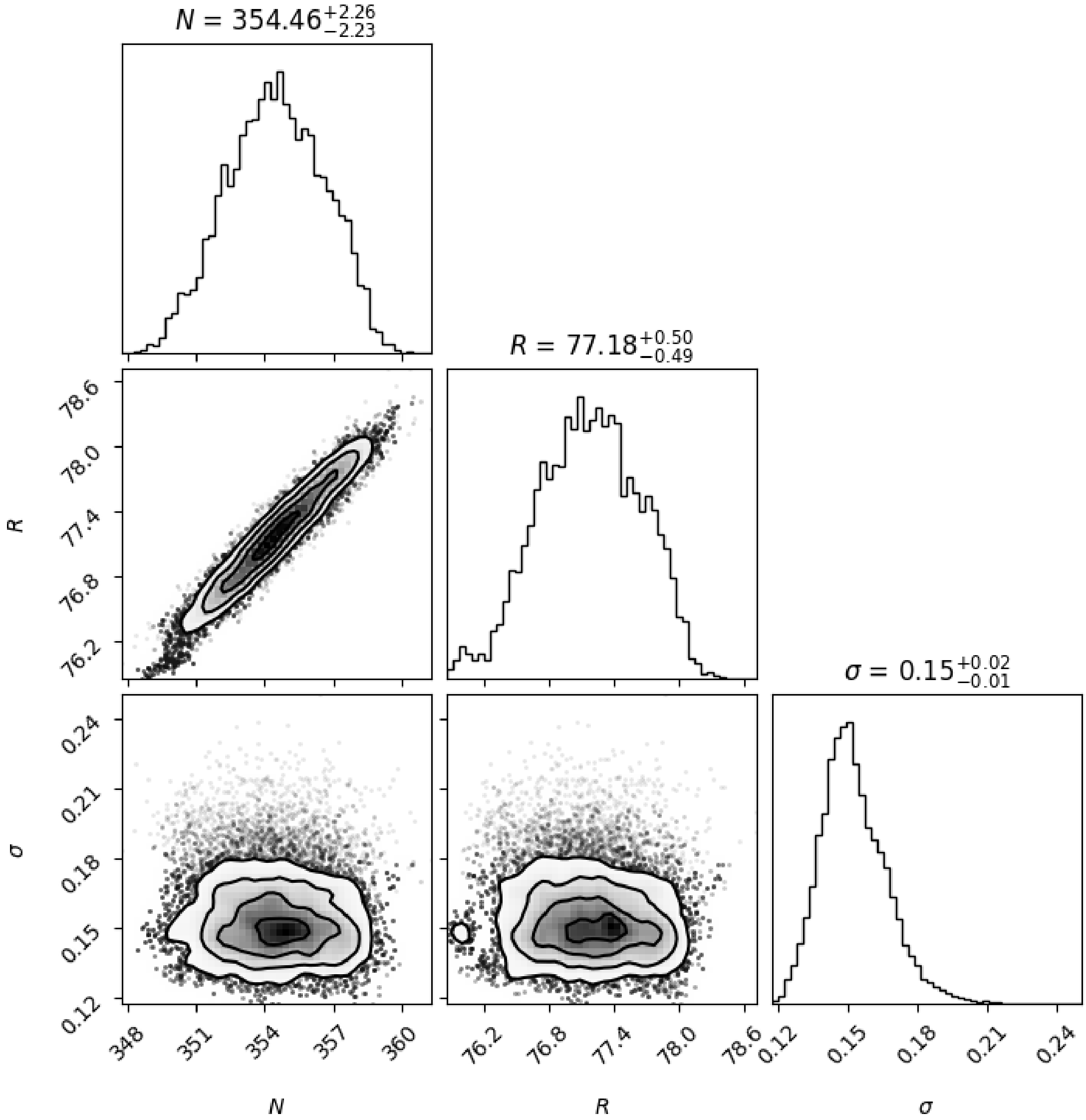

Figure 14: Corner plot from nested sampling for the isotropic model

## Declaration of Use of AI Tools

I declare that all ideas, methods, and approaches used in this project were conceived and developed independently through my own research, understanding and engagement with lecture and supervision material. The work presented here is entirely my own.

The following AI tools were used for support purposes:

- **ChatGPT:** Used to refine code snippets, assist with LaTeX formatting, and improve the clarity and structure. All content generated by this tool was critically evaluated and adapted.
- **GitHub Copilot:** Used to suggest coding snippets and assist the implementation of functions. All suggested code was reviewed, modified, and tested.